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Package structure and manufacturing method thereof

Abstract

A package structure includes a circuit board, a package substrate, an electronic/photonic assembly, a film redistribution layer, a heat dissipation assembly, and an optical fiber assembly. The package substrate is disposed on the circuit board and electrically connected to the circuit board. The electronic/photonic assembly includes an ASIC assembly, an EIC assembly, and a PIC assembly. The EIC assembly and the PIC assembly are stacked and disposed on the package substrate and electrically connected to the package substrate via the film redistribution layer. An orthographic projection of the EIC assembly on the film redistribution layer is overlapped with an orthographic projection of the PIC assembly on the film redistribution layer. The heat dissipation assembly is disposed on the electronic/photonic assembly. The optical fiber assembly is disposed on the package substrate and optically connected to the PIC assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation-in-part application of and claims the priority benefit of a prior application Ser. No. 17/835,990, filed on Jun. 9, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The invention relates to a semiconductor structure and a manufacturing method thereof, and particularly relates to a package structure and a manufacturing method thereof.

Description of Related Art

(2) In recent years, high-performance computing (HPC) has become more popular and widely used in advanced network and server applications, especially for artificial intelligence (AI)-related products that require high data rates, increasing bandwidth, and decreasing latency. People have more expectations and requirements for high density (HD) package substrates used in package structures including high performance computing (HPC). For example, the line width and the line spacing of the metal layer are getting smaller, and the thickness of the dielectric layer of the redistribution circuit layer is getting thinner. However, the current build-up package substrate may

not meet the above requirements. Therefore, in order to meet the above requirements, the industry proposes to place an application-specific integrated circuit (ASIC) assembly, an electronic integrated circuit (EIC) assembly, and a photonic integrated circuit (PIC) assembly side-by-side on a film redistribution layer to be electrically connected to each other via the film redistribution layer. However, since the ASIC assembly, the EIC assembly, and the PIC assembly are placed side-by-side, the area of the desired film redistribution layer is larger, and the area of the package substrate cannot be reduced. Furthermore, due to of the above arrangement, the electrical transmission path is farther away, and better electrical performance may not be achieved. Moreover, the industry further proposes to add a through-silicon via (TSV)-interposer on top of a build-up package substrate, but the price of the TSV-interposer is very high.

SUMMARY OF THE INVENTION

(3) The invention provides a package structure that may solve the issues of the prior art and has lower cost.

(4) The invention further provides a manufacturing method of a package structure for manufacturing the above package structure.

(5) A package structure of the invention includes a circuit board, a package substrate, an electronic/photonic assembly, a film redistribution layer (or an organic interposer), a heat dissipation assembly, and an optical fiber assembly. The package substrate is disposed on the circuit board and electrically connected to the circuit board. The electronic/photonic assembly includes an application-specific integrated circuit (ASIC) assembly, an electronic integrated circuit (EIC) assembly, and a photonic integrated circuit (PIC) assembly. The ASIC assembly is disposed on the package substrate and electrically connected to the package substrate. The EIC assembly and the PIC assembly are stacked and disposed on the package substrate and electrically connected to the package substrate. The film redistribution layer (or an organic interposer) is disposed between the EIC assembly and the PIC assembly. The EIC assembly and the PIC assembly are electrically connected to the package substrate via the film redistribution layer. An orthographic projection of the EIC assembly on the film redistribution layer is overlapped with an orthographic projection of the PIC assembly on the film redistribution layer. The heat dissipation assembly is disposed on the electronic/photonic assembly. The optical fiber assembly is disposed on the package substrate and optically connected to the PIC assembly.

(6) In an embodiment of the invention, the PIC assembly is located between the film redistribution layer and the package substrate.

(7) In an embodiment of the invention, the package substrate includes an accommodating groove, and a portion of the PIC assembly is located in the accommodating groove.

(8) In an embodiment of the invention, the heat dissipation assembly includes a first heat spreader, a second heat spreader, a first thermoelectric cooling fin, and a second thermoelectric cooling fin. The first heat spreader is disposed on the ASIC assembly. The second heat spreader is disposed on the PIC assembly. The first thermoelectric cooling fin is disposed on the first heat spreader and located between the first heat spreader and the ASIC assembly. The second thermoelectric cooling fin is disposed on the second heat spreader and located between the second heat spreader and the PIC assembly.

(9) In an embodiment of the invention, the heat dissipation assembly further includes a plurality of first thermal interface materials and a plurality of second thermal interface materials. The plurality of first thermal interface materials are respectively disposed between the first thermoelectric cooling fin and the first heat spreader, and between the first thermoelectric cooling fin and the ASIC assembly. The plurality of second thermal interface materials are respectively disposed between the second thermoelectric cooling fin and the second heat spreader, and between the second thermoelectric cooling fin and the EIC assembly.

(10) In an embodiment of the invention, the EIC assembly is located between the film redistribution layer and the package substrate.

(11) In an embodiment of the invention, the package substrate includes an accommodating groove, and a portion of the EIC assembly is located in the accommodating groove.

(12) In an embodiment of the invention, the heat dissipation assembly includes a first heat spreader, a second heat spreader, a first thermoelectric cooling fin, and a second thermoelectric cooling fin. The first heat spreader is disposed on the ASIC assembly. The second heat spreader is disposed on the EIC assembly. The first thermoelectric cooling fin is disposed on the first heat spreader and located between the first heat spreader and the ASIC assembly. The second thermoelectric cooling fin is disposed on the second heat spreader and located between the second heat spreader and the EIC assembly.

(13) In an embodiment of the invention, the heat dissipation assembly further includes a plurality of first thermal interface materials and a plurality of second thermal interface materials. The plurality of first thermal interface materials are respectively disposed between the first thermoelectric cooling fin and the first heat spreader, and between the first thermoelectric cooling fin and the ASIC assembly. The plurality of second thermal interface materials are respectively disposed between the second thermoelectric cooling fin and the second heat spreader, and between the second thermoelectric cooling fin and the PIC assembly.

(14) In an embodiment of the invention, the package structure further includes a plurality of first conductive members, a plurality of second conductive members, and a plurality of third conductive members. The first conductive members are disposed between the package substrate and the circuit board, wherein the package substrate is electrically connected to the circuit board via the first conductive members. The second conductive members are disposed between the film redistribution layer and the package substrate, between the film redistribution layer and the PIC assembly, and between the film redistribution layer and the EIC assembly. The EIC assembly and the PIC assembly are electrically connected to the film redistribution layer via the second conductive members. The film redistribution layer is electrically connected to the package substrate via the second conductive members. The third conductive members are disposed between the ASIC assembly and the package substrate. The ASIC assembly is electrically connected to the package substrate via the third conductive members. Each of the first conductive members, each of the second conductive members, and each of the third conductive members respectively include a solder ball or a C4 bump.

(15) In an embodiment of the invention, the package structure further includes an underfill disposed between the ASIC assembly and the package substrate and covering the third conductive members.

(16) In an embodiment of the invention, the optical fiber assembly includes an optical fiber connector, an optical coupler, and an optical fiber cable. The optical fiber connector is disposed on the package substrate and electrically connected to the package substrate. The optical fiber cable passes through the optical fiber connector and is electrically connected to the PIC assembly via the optical coupler.

(17) A manufacturing method of a package structure of the invention includes the following steps. A wafer bump of an ASIC assembly wafer is provided and the wafer bump of the ASIC assembly wafer is singulated into an individual ASIC assembly. An organic interposer is manufactured using a temporary glass carrier and the organic interposer is singulated into an individual organic interposer. A wafer bump of an EIC assembly wafer and a wafer bump of a PIC assembly wafer are provided and the wafer bump of the EIC assembly wafer and the wafer bump of the PIC assembly wafer are singulated into an individual EIC assembly and an individual PIC assembly. The individual EIC assembly or the individual PIC assembly is attached on the individual organic interposer carrying the temporary glass carrier. The temporary glass carrier is removed and the individual EIC assembly or the individual PIC assembly is attached to a bottom of the organic interposer, and then a copper pillar and a solder cap are electroplated. A three-dimensional stack of the ASIC assembly and the EIC assembly or the PIC assembly is connected on a package substrate

with/without an accommodating groove, and then the package substrate is connected on a circuit board using a solder ball. A thermal management system is applied on the ASIC assembly and the EIC assembly or the PIC assembly.

(18) Based on the above, in the design of the package structure of the invention, the ASIC assembly of the electronic/photonic assembly is disposed on the package substrate and electrically connected to the package substrate, and the EIC assembly and the PIC assembly of the electronic/photonic assembly are stacked and disposed on the package substrate and electrically connected to the package substrate via the film redistribution layer. Compared with the build-up package substrate or the TSV interposer in the prior art, the package structure of the invention not only meets people's expectations and requirements for a high-density package structure, but also has lower cost, high performance, and may adopt a smaller package substrate.

(19) In order to make the aforementioned features and advantages of the disclosure more comprehensible, embodiments accompanied with figures are described in detail below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic cross-sectional view of a package structure according to an embodiment of the invention.

(2) FIG. 2 is a schematic cross-sectional view of a package structure according to another embodiment of the invention.

(3) FIG. 3 is a schematic cross-sectional view of a package structure according to another embodiment of the invention.

(4) FIG. 4 is a schematic cross-sectional view of a package structure according to another embodiment of the invention.

(5) FIG. 5 is a schematic flowchart of a manufacturing method of a package structure according to an embodiment of the invention.

DESCRIPTION OF THE EMBODIMENTS

(6) The embodiments of the invention may be understood together with the drawings, and the drawings of the invention are also regarded as a part of the disclosure. It should be understood that the drawings of the invention are not drawn to scale and, in fact, the dimensions of elements may be arbitrarily expanded or reduced in order to clearly represent the features of the invention.

(7) FIG. 1 is a schematic cross-sectional view of a package structure according to an embodiment of the invention. Referring to FIG. 1, in the present embodiment, a package structure **100a** includes a circuit board **110**, a package substrate **120a**, an electronic/photonic assembly **130a**, a film redistribution layer **140**, a heat dissipation assembly **150a**, and an optical fiber assembly **160**. The package substrate **120a** is disposed on the circuit board **110** and electrically connected to the circuit board **110**. The electronic/photonic assembly **130a** includes an application-specific integrated circuit (ASIC) assembly **132**, an electronic integrated circuit (EIC) assembly **134a**, and a photonic integrated circuit (PIC) assembly **136a**. The ASIC assembly **132** is disposed on the package substrate **120a** and electrically connected to the package substrate **120a**. The EIC assembly **134a** and the PIC assembly **136a** are stacked and disposed on the package substrate **120a** and electrically connected to the package substrate **120a**. The film redistribution layer **140** is disposed between the EIC assembly **134a** and the PIC assembly **136a**. The EIC assembly **134a** and the PIC assembly **136a** are electrically connected to the package substrate **120a** via the film redistribution layer **140**. The orthographic projection of the EIC assembly **134a** on the film redistribution layer **140** is overlapped with the orthographic projection of the PIC assembly **136a** on the film redistribution layer **140**. The heat dissipation assembly **150a** is disposed on the electronic/photonic assembly **130a**. The optical fiber assembly **160** is disposed on the package substrate **120a** and optically

connected to the PIC assembly **136a**. In an embodiment, an organic interposer may also be used to replace the film redistribution layer **140**, which is still within the scope of the invention.

(8) In short, the ASIC assembly **132** of the present embodiment is disposed on the package substrate **120a** and electrically connected to the package substrate **120a**, and the EIC assembly **134a** and the PIC assembly **136a** are stacked and disposed on the package substrate **120a** and electrically connected to the package substrate **120a** via the film redistribution layer **140**.

Compared with the build-up package substrate or the TSV interposer in the prior art, the package structure **100a** of the present embodiment not only may meet people's expectations and requirements for high-density package structures, the cost of the film redistribution layer **140** is lower than that of the TSV interposer in the prior art, and therefore the cost may be lower, and the film redistribution layer **140** is disposed between the EIC assembly **134a** and the PIC assembly **136a**, thus effectively reducing the area of the package substrate **120a**, so that a smaller package substrate **120a** may be adopted, and better electrical performance is achieved.

(9) Specifically, referring further to FIG. **1**, in the present embodiment, the PIC assembly **136a** is embodied between the film redistribution layer **140** and the package substrate **120a**. The film redistribution layer **140** includes a plurality of patterned circuit layers **142**, a plurality of dielectric layers **144**, a plurality of first pads **146**, a plurality of second pads **148**, a plurality of conductive blind vias **145**, and a solder resist layer **149**. The patterned circuit layers **142** and the dielectric layers **144** are alternately stacked, wherein the line width and the line pitch of the patterned circuit layers **142** are both, for example, 2 micrometers, 5 micrometers, or 10 micrometers, meaning that the patterned circuit layers **142** are thin circuit layers. Preferably, the wiring density of the film redistribution layer **140** is greater than the wiring density of the package substrate **120a**, and the wiring density of the package substrate **120a** is greater than the wiring density of the circuit board **110**. The material of the dielectric layer **144** may be, for example, organic material, glass, or ceramic, but not limited thereto. The surface of the first pads **146** is exposed and aligned with the surface of the dielectric layer **144** closest to the electronic/photonic assembly **130a**. The second pads **148** are directly electrically connected to the patterned circuit layers **142**, wherein the surface of the second pads **148** is exposed and aligned with the surface of the solder resist layer **149**. The conductive blind vias **145** pass through the dielectric layers **144** and are electrically connected between the patterned circuit layers **142** and between the patterned circuit layers **142** and the first pads **146**.

(10) Furthermore, the heat dissipation assembly **150a** of the present embodiment includes a first heat spreader **152**, a second heat spreader **154a**, a first thermoelectric cooling fin **156**, and a second thermoelectric cooling fin **158a**. The first heat spreader **152** is disposed on the ASIC assembly **132**. The second heat spreader **154a** is disposed on the EIC assembly **134a**. The first thermoelectric cooling fin **156** is disposed on the first heat spreader **152** and located between the first heat spreader **152** and the ASIC assembly **132**. The second thermoelectric cooling fin **158a** is disposed on the second heat spreader **154a** and located between the second heat spreader **154a** and the EIC assembly **134a**.

(11) Furthermore, the heat dissipation assembly **150a** of the present embodiment further includes a plurality of first thermal interface materials **151** and **153** and a plurality of second thermal interface materials **157** and **159**. The first thermal interface material **151** is disposed between the first thermoelectric cooling fin **156** and the first heat spreader **152**, and the first thermal interface material **153** is disposed between the first thermoelectric cooling fin **156** and the ASIC assembly **132**. The second thermal interface material **157** is disposed between the second thermoelectric cooling fin **158a** and the second heat spreader **154a**, and the second thermal interface material **159** is disposed between the second thermoelectric cooling fin **158a** and the EIC assembly **134a**.

(12) Furthermore, the package structure **100a** of the present embodiment further includes a plurality of first conductive members **170**, a plurality of second conductive members **175**, and a plurality of third conductive members **180**. The first conductive members **170** are disposed between

the package substrate **120a** and the circuit board **110**, wherein the package substrate **120a** is electrically connected to the circuit board **110** via the first conductive members **170**. The second conductive members **175** are disposed between the film redistribution layer **140** and the package substrate **120a**, between the film redistribution layer **140** and the PIC assembly **136a**, and between the film redistribution layer **140** and the EIC assembly **134a**. The EIC assembly **134a** and the PIC assembly **136a** are electrically connected to the film redistribution layer **140** via the second conductive members **175**. The film redistribution layer **140** is electrically connected to the package substrate **120a** via the second conductive members **175**. The third conductive members **180** are disposed between the ASIC assembly **132** and the package substrate **120a**. The ASIC assembly **132** is electrically connected to the package substrate **120a** via the third conductive members **180**. Each of the first conductive members **170**, each of the second conductive members **175**, and each of the third conductive members **180** is, for example, a solder ball or a C4 bump respectively, which is not limited here. In addition, the package structure **100a** of the present embodiment further includes an underfill **185** disposed between the ASIC assembly **132** and the package substrate **120a** and covering the third conductive members **180**.

(13) It should be mentioned that, since the thermal expansion coefficient of the film redistribution layer **140** is comparable to those of the package substrate **120a**, the PIC assembly **136a**, and the EIC assembly **134a**, there is no underfill needed between the film redistribution layer **140** and the package substrate **120a**, between the film redistribution layer **140** and the PIC assembly **136a**, and between the film redistribution layer **140** and the EIC assembly **134a**.

(14) In addition, please refer to FIG. 1 again, the optical fiber assembly **160** of the present embodiment includes an optical fiber connector **162**, an optical coupler **164**, and an optical fiber cable **166**. The optical fiber connector **162** is disposed on the package substrate **120a** and electrically connected to the package substrate **120a**. The optical fiber cable **166** passes through the optical fiber connector **162** and is electrically connected to the PIC assembly **136a** via the optical coupler **164**.

(15) FIG. 5 is a schematic flowchart of a manufacturing method of a package structure according to an embodiment of the invention. Please refer to FIG. 1 and FIG. 5 at the same time. In the manufacturing process of the package structure **100a**, first, step **S11**, a wafer bump of an ASIC assembly wafer is provided and the wafer bump of the ASIC assembly wafer is singulated into an individual ASIC assembly; step **S12**, an organic interposer (or a film redistribution layer) is manufactured using a temporary glass carrier and the organic interposer is singulated into an individual organic interposer; and step **S13**, a wafer bump of an EIC assembly wafer and a wafer bump of a PIC assembly wafer are provided and the wafer bump of the EIC assembly wafer and the wafer bump of the PIC assembly wafer are singulated into an individual EIC assembly and an individual PIC assembly. The above step **S11**, step **S12**, and step **S13** are in no order.

(16) Next, step **S20**, the individual EIC assembly **134a** or the individual PIC assembly **136a** is attached on the individual organic interposer (or the film redistribution layer **140**) carrying the temporary glass carrier.

(17) Next, step **S30**, the temporary glass carrier is removed and the individual EIC assembly **134a** or the individual PIC assembly **136a** is attached to a bottom of the organic interposer (or the film redistribution layer **140**), and then a copper pillar and a solder cap are electroplated. The EIC assembly **134a** and the PIC assembly **136a** are respectively disposed on two opposite sides of the film redistribution layer **140** to define an electronic module E. At this time, the orthographic projection of the EIC assembly **134a** on the film redistribution layer **140** is overlapped with the orthographic projection of the PIC assembly **136a** on the film redistribution layer **140**.

(18) Then, step **S40**, a three-dimensional stack of the ASIC assembly **132** and the EIC assembly **134a** or the PIC assembly **136a** is connected on a package substrate **120a** with/without an accommodating groove, and then the package substrate **120a** is connected on the circuit board **110** using a solder ball (such as the first conductive members **170**). The ASIC assembly **132** and the

electronic module E are disposed on the package substrate **120a**, wherein the ASIC assembly **132** and the electronic module E are respectively electrically connected to the package substrate **120a**. Specifically, the ASIC assembly **132** may be bonded on the package substrate **120a** and electrically connected to the package substrate **120a** via the third conductive members **180** formed between the ASIC assembly **132** and the package substrate **120a**, wherein each of the third conductive members **180** may be, for example, a solder ball or a C4 bump. The electronic module E may be bonded on the package substrate **120a** and electrically connected to the package substrate **120a** via the second conductive members **175** formed between the film redistribution layer **140** and the package substrate **120a**.

(19) Lastly, step S50, a thermal management system is applied on the ASIC assembly **132** and the EIC assembly **134a** or the PIC assembly **136a**. That is, the heat dissipation assembly **150a** is disposed on the electronic/photonic assembly **130a**. Immediately afterwards, the optical fiber assembly **160** is disposed on the package substrate **120a**, wherein the optical fiber assembly **160** is optically connected to the PIC assembly **136a**. At this point, the manufacture of the package structure **100a** is completed.

(20) It should be mentioned that, the embodiments below use the same reference numerals and portions of the content from previous embodiments. Specifically, the same reference numerals are used to represent the same or similar elements, and the descriptions for the same techniques are omitted. The omitted portions are as described in the embodiments above and are not repeated in the embodiments below.

(21) FIG. 2 is a schematic cross-sectional view of a package structure according to another embodiment of the invention. Please refer to FIG. 1 and FIG. 2 at the same time. A package structure **100b** of the present embodiment is similar to the package structure **100a** of FIG. 1, and the difference between the two is: in the present embodiment, the thickness of a PIC assembly **136b** is greater than the thickness of the PIC assembly **136a** of the above embodiment, so a package substrate **120b** may include an accommodating groove **122**, and a portion of the PIC assembly **136b** may be located in the accommodating groove **122**. At this time, an EIC assembly **134b** and the PIC assembly **136b** of an electronic/photonic assembly **130b** are still stacked and disposed on the package substrate **120b**, and the orthographic projection of the EIC assembly **134b** on the film redistribution layer **140** is completely overlapped with the orthographic projection of the PIC assembly **136b** on the film redistribution layer **140**.

(22) In terms of manufacturing process, please refer to FIG. 2 and FIG. 5 at the same time. The manufacturing method of the package structure **100b** of the present embodiment is similar to the manufacturing method of the package structure **100a**. The difference between the two is: in the present embodiment, in step S13, the package substrate **120b** formed with the accommodating groove **122** may be directly provided to correspond to the subsequent PIC assembly **136b**.

(23) FIG. 3 is a schematic cross-sectional view of a package structure according to another embodiment of the invention. Please refer to FIG. 1 and FIG. 3 at the same time. A package structure **100c** of the present embodiment is similar to the package structure **100a** of FIG. 1, and the difference between the two is: in the present embodiment, an EIC assembly **134c** of an electronic/photonic assembly **130c** is located between the film redistribution layer **140** and the package substrate **120a**. Furthermore, a second heat spreader **154b** of a heat dissipation assembly **150b** is disposed on a PIC assembly **136c**. A second thermoelectric cooling fin **158b** of the heat dissipation assembly **150b** is disposed on the second heat spreader **154b** and located between the second heat spreader **154b** and the PIC assembly **136c**. Moreover, the second thermal interface material **157** of the heat dissipation assembly **150b** is disposed between the second thermoelectric cooling fin **158b** and the second heat spreader **154b**, and the second thermal interface material **157** of the heat dissipation assembly **150b** is disposed between the second thermoelectric cooling chip **158b** and the PIC assembly **136c**.

(24) FIG. 4 is a schematic cross-sectional view of a package structure according to another

embodiment of the invention. Please refer to FIG. 3 and FIG. 4 at the same time. A package structure **100d** of the present embodiment is similar to the package structure **100c** of FIG. 3, and the difference between the two is: in the present embodiment, the thickness of an EIC assembly **136d** is greater than the thickness of the EIC assembly **136c** of the above embodiment, so the package substrate **120b** may include an accommodating groove **122**, and a portion of the EIC assembly **136d** may be located in the accommodating groove **122**. At this time, an EIC assembly **134d** and the PIC assembly **136d** of an electronic/photonic assembly **130d** are still stacked and disposed on the package substrate **120b**, and the orthographic projection of the PIC assembly **134d** on the film redistribution layer **140** is completely overlapped with the orthographic projection of the EIC assembly **136d** on the film redistribution layer **140**.

(25) Based on the above, in the design of the package structure of the invention, the ASIC assembly of the electronic/photonic assembly is disposed on the package substrate and electrically connected to the package substrate, and the EIC assembly and the PIC assembly of the electronic/photonic assembly are stacked and disposed on the package substrate and electrically connected to the package substrate via the film redistribution layer. Compared with the build-up package substrate or the TSV interposer in the prior art, the package structure of the invention not only meets people's expectations and requirements for a high-density package structure, but also has lower cost, high performance, and may adopt a smaller package substrate.

(26) Although the invention has been described with reference to the above embodiments, it will be apparent to one of ordinary skill in the art that modifications to the described embodiments may be made without departing from the spirit of the disclosure. Accordingly, the scope of the disclosure is defined by the attached claims not by the above detailed descriptions.

Claims

1. A package structure, comprising: a circuit board; a package substrate disposed on the circuit board and electrically connected to the circuit board; an electronic/photonic assembly comprising an application specific integrated circuit (ASIC) assembly, an electronic integrated circuit (EIC) assembly, and a photonic integrated circuit (PIC) assembly, wherein the ASIC assembly is disposed on the package substrate and electrically connected to the package substrate, and the EIC assembly and the PIC assembly are stacked and disposed on the package substrate and electrically connected to the package substrate; a film redistribution layer disposed only between the EIC assembly and the PIC assembly, wherein the EIC assembly and the PIC assembly are electrically connected to the package substrate via the film redistribution layer, and an orthographic projection of the EIC assembly on the film redistribution layer is overlapped with an orthographic projection of the PIC assembly on the film redistribution layer; a heat dissipation assembly disposed on the electronic/photonic assembly; and an optical fiber assembly disposed on the package substrate and optically connected to the PIC assembly.
2. The package structure of claim 1, wherein the PIC assembly is located between the film redistribution layer and the package substrate.
3. The package structure of claim 2, wherein the package substrate comprises an accommodating groove, and a portion of the PIC assembly is located in the accommodating groove.
4. The package structure of claim 2, wherein the heat dissipation assembly comprises: a first heat spreader disposed on the ASIC assembly; a second heat spreader disposed on the PIC assembly; a first thermoelectric cooling fin disposed on the first heat spreader and located between the first heat spreader and the ASIC assembly; and a second thermoelectric cooling fin disposed on the second heat spreader and located between the second heat spreader and the PIC assembly.
5. The package structure of claim 4, wherein the heat dissipation assembly further comprises: a plurality of first thermal interface materials respectively disposed between the first thermoelectric cooling fin and the first heat spreader, and between the first thermoelectric cooling fin and the

ASIC assembly; and a plurality of second thermal interface materials respectively disposed between the second thermoelectric cooling fin and the second heat spreader, and between the second thermoelectric cooling fin and the EIC assembly.

6. The package structure of claim 1, wherein the EIC assembly is located between the film redistribution layer and the package substrate.

7. The package structure of claim 6, wherein the package substrate comprises an accommodating groove, and a portion of the EIC assembly is located in the accommodating groove.

8. The package structure of claim 6, wherein the heat dissipation assembly comprises: a first heat spreader disposed on the ASIC assembly; a second heat spreader disposed on the EIC assembly; a first thermoelectric cooling fin disposed on the first heat spreader and located between the first heat spreader and the ASIC assembly; and a second thermoelectric cooling fin disposed on the second heat spreader and located between the second heat spreader and the EIC assembly.

9. The package structure of claim 8, wherein the heat dissipation assembly further comprises: a plurality of first thermal interface materials respectively disposed between the first thermoelectric cooling fin and the first heat spreader, and between the first thermoelectric cooling fin and the ASIC assembly; and a plurality of second thermal interface materials respectively disposed between the second thermoelectric cooling fin and the second heat spreader, and between the second thermoelectric cooling fin and the PIC assembly.

10. The package structure of claim 1, further comprising: a plurality of first conductive members disposed between the package substrate and the circuit board, wherein the package substrate is electrically connected to the circuit board via the first conductive members; a plurality of second conductive members disposed between the film redistribution layer and the package substrate, between the film redistribution layer and the PIC assembly, and between the film redistribution layer and the EIC assembly, wherein the EIC assembly and the PIC assembly are electrically connected to the film redistribution layer via the second conductive members, and the film redistribution layer is electrically connected to the package substrate via the second conductive members; and a plurality of third conductive members disposed between the ASIC assembly and the package substrate, wherein the ASIC assembly is electrically connected to the package substrate via the third conductive members, and each of the first conductive members, each of the second conductive members, and each of the third conductive members respectively comprise a solder ball or a C4 bump.

11. The package structure of claim 10, further comprising: an underfill disposed between the ASIC assembly and the package substrate and covering the third conductive members.

12. The package structure of claim 1, wherein the optical fiber assembly comprises an optical fiber connector, an optical coupler, and an optical fiber cable, the optical fiber connector is disposed on the package substrate and electrically connected to the package substrate, and the optical fiber cable passes through the optical fiber connector and is electrically connected to the PIC assembly via the optical coupler.

13. A manufacturing method of a package structure, comprising: providing a wafer bump of an ASIC assembly wafer and singulating the wafer bump of the ASIC assembly wafer into an individual ASIC assembly; manufacturing an organic interposer using a temporary glass carrier and singulating the organic interposer into an individual organic interposer; providing a wafer bump of an EIC assembly wafer and a wafer bump of a PIC assembly wafer and singulating the wafer bump of the EIC assembly wafer and the wafer bump of the PIC assembly wafer into an individual EIC assembly and an individual PIC assembly; attaching the individual EIC assembly or the individual PIC assembly on the individual organic interposer carrying the temporary glass carrier; removing the temporary glass carrier and attaching the individual EIC assembly or the individual PIC assembly to a bottom of the organic interposer, and then electroplating a copper pillar and a solder cap; connecting a three-dimensional stack of the ASIC assembly and the EIC assembly or the PIC assembly on a package substrate with/without an accommodating groove, and then connecting the

package substrate on a circuit board using a solder ball; and applying a thermal management system on the ASIC assembly and the EIC assembly or the PIC assembly.
